

## AIRSTREET · REQUIREMENTS

# AirStreet requirements

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## DOCUMENT CONTROL

|             |                                        |         |                |
|-------------|----------------------------------------|---------|----------------|
| Document ID | AST-REQ-001                            | Version | 0.8            |
| Status      | <b>Draft</b>                           | Date    | 2026-10-02     |
| Author      | Amish Chadha                           | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept          |         |                |
| Repository  | github.com/BoujeeEnjinia1701/airstreet |         |                |
| Source      | docs/03-requirements.md @ uncommitted  |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                                                                             |
|---------|------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.8     | 2026-10-02 | Amish Chadha | Status from AST-CAL-001 v0.6 with the four lightening steps carried into the design: R13 at risk to met on paper (3.78 kg with the sun shield); R15 over the target by \$17.50; R5 figures updated |
| 0.7     | 2026-10-02 | Amish Chadha | R13 mass limit set at 4.0 kg including FieldNode's sun shield (AST-DDR-003 A1, accepted 2026-10-02); R13 now at risk                                                                               |
| 0.6     | 2026-10-01 | Amish Chadha | Budget treated as a value-engineering target                                                                                                                                                       |
| 0.5     | 2026-09-30 | Amish Chadha | Status from AST-CAL-001 v0.3 after the design was made constructable (AST-DDR-003); R13 and R15 not met, proposed to Amish                                                                         |
| 0.4     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002)                                                                                                                                                        |
| 0.3     | 2026-09-25 | Amish Chadha | TRL 3 status from AST-CAL-001; R15 redefined as the sensor-head cost (AST-DDR-001 D1); R6, R8 and R11 rest on adopted choices                                                                      |

|     |            |              |                                                                          |
|-----|------------|--------------|--------------------------------------------------------------------------|
| 0.2 | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with status against the concept |
| 0.1 | 2026-09-25 | Amish Chadha | Initial scaffold                                                         |

# AirStreet requirements

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These requirements are checked by calculation in [AST-CAL-001](#) v0.6. Targets remain proposals for review with the first co-design partner. On 2026-09-25 Amish accepted the recommendations in [AST-DDR-001](#) and [AST-DDR-002](#). As a result R3 is withdrawn and kept as a research question, R8, R12, R13, R15 and R16 are restated, and R15 uses the \$285 `budget_usd` . On 2026-09-30 the design was made constructable ([AST-DDR-003](#)), which raised the mass and the sensor-head cost: R15 (cost) is **over the value-engineering target**, now by \$17.50. On 2026-10-02 Amish accepted the design for construction and set R13's mass limit at 4.0 kg including FieldNode's sun shield (AST-DEC-001, decision of 2026-10-02 on mass), with the four lightening steps carried into the design. With them the node weighs 3.78 kg with the sun shield, so R13 is met on paper with 0.22 kg of margin (it was at risk, on its limit, before the lightening). Five are at risk, five are met on paper, four are met by design, one is over the value-engineering target and one is withdrawn.

Table 1. Requirements.

| ID  | Requirement                                            | Target                                                                                                                                                                           | Verification (TRL 3 or later)                                                | Status at TRL 3 (AST-CAL-001)                                                                                                                                         |
|-----|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1  | Measure PM2.5 at street level                          | 0 to 500 $\mu\text{g}/\text{m}^3$ ; hourly mean after humidity correction within $\pm 5 \mu\text{g}/\text{m}^3$ or $\pm 30 \%$ (whichever is larger) of a collocated reference   | CalRig chamber check, then field collocation                                 | At risk: within the bound up to 80 % RH; outside it above 85 % RH, where water uptake makes the correction uncertain (AST-CAL-001 E)                                  |
| R2  | Measure NO2 at street level                            | 0 to 200 ppb (about 0 to 380 $\mu\text{g}/\text{m}^3$ ); hourly mean absolute error 5 ppb or less against a reference station after field calibration                            | Field collocation with a chemiluminescence reference station                 | At risk: hourly MAE about 3.9 ppb from a budget of literature and assumed terms (AST-CAL-001 C)                                                                       |
| R3  | Resolve NO2 near the WHO annual guideline              | Withdrawn as a requirement (AST-DDR-002). Kept as a research question: can the annual mean be resolved to 2 $\mu\text{g}/\text{m}^3$ at 10 $\mu\text{g}/\text{m}^3$ ?            | Collocation statistics (research)                                            | Withdrawn. Annual uncertainty about 4.2 $\mu\text{g}/\text{m}^3$ ( $1\sigma$ ); AirStreet ranks streets and checks the EU limit (R4) and makes no WHO guideline claim |
| R4  | Distinguish streets that exceed the EU NO2 limit value | Classify annual means above or below 40 $\mu\text{g}/\text{m}^3$ with 90 % confidence                                                                                            | Collocation statistics                                                       | At risk: decisive only for annual means below 34.6 or above 45.4 $\mu\text{g}/\text{m}^3$                                                                             |
| R5  | Measure temperature and humidity for corrections       | $\pm 0.5 \text{ }^\circ\text{C}$ and $\pm 3 \%$ RH inside the shield; radiation error 1 $^\circ\text{C}$ or less in full sun at 1 m/s wind                                       | CalRig chamber check; side-by-side field test against an aspirated reference | At risk: SHT45 meets the sensor figures; shield error 0.42 to 0.87 K at 1 m/s, which reads 4.1 % RH low at 85 % RH                                                    |
| R6  | Calibrate with a stated method                         | T, RH and PM checked on CalRig before deployment; NO2 collocated with a reference station for at least 14 days before deployment and at least every 6 months                     | Calibration record per node                                                  | Met by design (AST-DDR-001 D3)                                                                                                                                        |
| R7  | Keep raw data for recalibration                        | Send raw NO2 working and auxiliary electrode signals, PM, T and RH with a calibration version tag                                                                                | Firmware and data schema review                                              | Met by design                                                                                                                                                         |
| R8  | Report often enough for street maps                    | One record every 5 min through a private gateway (TwinKit by default); one every 15 min on The Things Network; hourly means published within 15 min; 7 days of store and forward | Timing, airtime and storage calculation                                      | Met on paper: 5 min records on the private gateway; at 15 min TTN fair use is met up to SF9; 7 days in 40.3 kB                                                        |
| R9  | Fit the FieldNode power budget                         | Sensor average 115 mW or less; energy neutral at the FieldNode design sun hours                                                                                                  | Power budget calculation                                                     | Met on paper: 87.0 mW design load with the 60 s PM run; 2.33 Wh/day drawn against 7.75 Wh stored in the worst month; 6.6 days without sun                             |
| R10 |                                                        |                                                                                                                                                                                  | Airtime calculation                                                          |                                                                                                                                                                       |

| ID  | Requirement                            | Target                                                                                                                                                                                                                | Verification (TRL 3 or later)          | Status at TRL 3 (AST-CAL-001)                                                                                                                                                                                                       |
|-----|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | Stay within radio duty-cycle rules     | LoRaWAN airtime below 1 % in EU868 sub-bands                                                                                                                                                                          |                                        | Met on paper: 0.082 % at SF9, 0.60 % at SF12                                                                                                                                                                                        |
| R11 | Mount on street poles without drilling | Poles 80 to 200 mm diameter; band clamps only; inlets between 1.5 and 4 m above ground                                                                                                                                | Design review                          | Met by design (inlet 3.0 m, AST-DDR-001 D7; 140° V-saddles and bands cut to length fit the range, AST-DDR-003)                                                                                                                      |
| R12 | Survive outdoors                       | Pod inlets face down behind insect mesh with drip lid; electronics in the FieldNode IP65 enclosure; -10 to 50 °C, with FieldNode's sun shield required at sites where ambient exceeds 45 °C                           | Design review, later field trial       | At risk: the FieldNode sun shield is now designed (FND-DDR-003) and fixes to AirStreet's adapter plates (AST-DDR-003); SPS30 recommended range 20 to 80 % RH; condensation and insects unverified                                   |
| R13 | Light and compact on the pole          | Mass 4.0 kg or less including FieldNode's sun shield (relaxed from 3.5 kg, AST-DDR-003 A1, 2026-10-02; the pole owner is asked to confirm the load); frontal area 0.15 m² or less (relaxed from 0.12 m², AST-DDR-002) | Constructable model; weighing at TRL 4 | Met on paper: 3.78 kg with FieldNode's sun shield (3.62 kg without) against 4.0 kg, 0.22 kg margin, with the four lightening steps in the design; frontal area 0.139 m²; the pole owner's confirmation of the load is still to come |
| R14 | Collect levels only                    | No images, audio or personal identifiers collected or sent                                                                                                                                                            | Design review                          | Met by design                                                                                                                                                                                                                       |
| R15 | Low cost                               | AirStreet sensor-head parts at or under the \$285 value-engineering target per node ( <code>budget_usd</code> , a hypothetical control target); FieldNode core excluded and costed in FieldNode                       | Priced BOM (AST-CAL-001 I)             | <b>Over the value-engineering target by \$17.50:</b> sensor head \$302.50; \$433.50 with the FieldNode core (AST-DDR-003)                                                                                                           |
| R16 | Serviceable in the field               | Pre-allocated sensor pod exchanged in 15 min at the pole with hand tools; no sensor-level swaps at the pole (AST-DDR-002)                                                                                             | Design review                          | Met on paper: about 14 min for a pod exchange; the exchanged pod is serviced and re-allocated at the reference site                                                                                                                 |

## ■ Assumptions

- NO2 conversions use 1 ppb = 1.88 µg/m³ at 25 °C and 101.325 kPa.
- The WHO guideline level for annual NO2 is 10 µg/m³ and for PM2.5 is 5 µg/m³, as cited by the European Environment Agency (EEA, 2024). The EU annual limit value for NO2 is 40 µg/m³ under Directive 2008/50/EC, which also sets sampling inlets between 1.5 m and 4 m above ground (EUR-Lex).
- FieldNode allows 115 mW average for sensors at its 5-day autonomy ceiling, with 100 mW proposed as its published design value (FND-CAL-001); AirStreet fits either.
- `budget_usd` in `project.yaml` is \$285 for the sensor head (AST-DDR-002).

- R1 and R2 targets are proposals for review, set with reference to published field results; they are not taken from a standard.